

CHERDYN, Russian Federation

WMO: 239140

Lat: 60.400N

Lon: 56.517E

Elev: 678

StdP: 14.34

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-19.9	-30.6	1.0	-25.9	-24.5	1.4	-19.9	20.7	14.9	18.6	16.0	3.6	320	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	82.9	67.8	79.4	66.0	75.8	63.5	69.3	80.2	67.4	76.9	65.4	73.7	4.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	96.0	74.6	63.6	90.2	73.0	61.8	84.6	70.8	33.8	80.4	32.2	76.7	30.6	73.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.4	16.2	14.4	DB	-30.7	87.0	7.6	2.7	-36.2	88.9	-40.7	90.5	-45.0	92.1	-50.5	94.1
(5)				WB	-30.9	72.2	7.6	2.8	-36.3	74.2	-40.8	75.8	-45.0	77.4	-50.5	79.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	5.5	8.2	22.6	35.7	48.7	59.2	64.2	57.7	47.7	34.7	19.2	9.5	(6)
(7)		DBStd	22.77	13.73	13.08	8.79	8.82	10.03	8.38	8.19	7.79	7.64	8.24	12.00	14.73	(7)
(8)		HDD50	6809	1379	1169	849	433	152	18	4	15	132	478	924	1256	(8)
(9)		HDD65	11273	1844	1589	1314	878	512	213	116	253	521	938	1374	1721	(9)
(10)		CDD50	1170	0	0	0	5	111	292	444	253	61	4	0	0	(10)
(11)		CDD65	161	0	0	0	0	6	37	91	26	1	0	0	0	(11)
(12)		CDH74	1126	0	0	0	0	47	257	645	172	5	0	0	0	(12)
(13)		CDH80	237	0	0	0	0	4	49	142	42	0	0	0	0	(13)
(14)	Wind	WSAvg	6.4	6.6	6.9	7.6	7.0	6.6	6.0	4.8	5.4	6.2	7.2	6.7	6.3	(14)
(15)	Precipitation	PrecAvg	28.0	2.0	1.4	1.4	1.8	2.1	2.6	3.3	2.9	2.6	2.8	2.7	2.4	(15)
(16)		PrecMax	38.0	3.7	2.7	4.1	4.0	4.9	6.7	7.4	6.4	4.8	5.8	6.2	4.4	(16)
(17)		PrecMin	18.9	0.3	0.0	0.1	0.6	0.4	0.9	0.6	0.9	0.7	0.6	0.1	0.2	(17)
(18)		PrecStd	4.4	0.7	0.6	0.8	0.9	1.0	1.5	1.5	1.3	1.1	1.1	1.1	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	33.2	44.8	66.2	79.3	85.0	86.7	85.4	72.7	59.1	44.6	35.0	(19)
(20)			MCWB	31.2	31.7	36.4	50.3	60.0	68.7	69.3	68.4	60.5	49.9	43.6	34.1	(20)
(21)		2%	DB	29.8	31.6	39.2	57.8	74.3	80.6	84.0	78.5	66.1	53.5	38.6	32.7	(21)
(22)			MCWB	29.2	30.8	33.5	46.3	57.3	65.9	68.5	66.4	57.2	48.3	37.6	32.2	(22)
(23)		5%	DB	26.8	29.4	36.0	52.2	69.7	76.6	80.9	73.7	62.1	49.5	34.8	30.9	(23)
(24)			MCWB	26.2	28.5	32.0	42.4	54.9	63.0	67.3	63.4	55.2	46.0	34.1	30.3	(24)
(25)		10%	DB	23.6	26.2	33.6	48.2	65.0	72.6	77.5	69.6	58.4	46.1	32.5	27.6	(25)
(26)			MCWB	23.0	25.4	30.9	40.5	52.9	61.1	65.8	61.4	53.5	44.0	32.0	27.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.6	32.4	37.8	51.8	62.8	70.4	72.1	70.0	62.3	52.5	43.2	34.5	(27)
(28)			MADB	31.9	32.7	42.4	61.2	73.8	82.3	83.4	81.7	70.2	55.7	44.2	34.8	(28)
(29)		2%	WB	29.3	30.9	34.9	47.1	59.6	67.2	70.1	67.3	58.6	49.1	37.6	32.2	(29)
(30)			MADB	29.8	31.5	37.8	55.8	70.0	77.3	81.1	77.2	64.3	52.6	38.3	32.5	(30)
(31)		5%	WB	26.3	28.5	32.9	43.8	56.8	64.8	68.5	64.6	56.2	46.4	34.2	30.2	(31)
(32)			MADB	26.7	29.3	35.1	50.3	66.5	73.8	78.4	71.8	60.4	49.1	34.7	30.8	(32)
(33)		10%	WB	23.0	25.5	31.2	41.2	54.1	62.7	66.8	62.4	54.1	43.9	32.2	27.0	(33)
(34)			MADB	23.6	26.2	33.4	47.1	63.8	70.8	75.5	68.3	57.8	46.1	32.5	27.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.7	12.0	13.4	16.2	15.5	15.4	13.1	11.2	7.3	7.5	9.2	(35)
(36)			MCDBR	9.7	8.9	13.0	19.3	21.7	19.9	19.3	18.1	16.3	12.2	6.8	7.5	(36)
(37)			MCWBR	10.0	8.9	9.2	11.6	10.7	9.6	8.6	8.3	8.8	8.1	6.9	7.8	(37)
(38)		5% WB	MCDDBR	10.1	9.0	10.8	17.5	19.4	18.4	17.9	16.7	14.3	11.1	6.6	7.8	(38)
(39)			MCWBR	10.3	9.1	8.6	11.3	11.3	9.6	9.2	8.5	8.8	8.5	6.8	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.272	0.305	0.349	0.375	0.374	0.372	0.401	0.396	0.357	0.326	0.280	0.257	(40)	
(41)		taud	1.969	2.012	1.982	2.074	2.353	2.415	2.332	2.327	2.396	2.301	2.057	1.921	(41)	
(42)		Ebn at Noon	138	206	236	257	270	272	259	251	241	205	136	91	(42)	
(43)		Edh at Noon	15	27	38	42	35	33	35	33	26	20	14	10	(43)	
(44)	All-Sky Solar Radiation	RadAvg	111	354	721	1150	1514	1613	1665	1166	656	264	101	58	(44)	
(45)		RadStd	16	75	101	171	164	148	165	142	72	36	25	9	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page